

Sardar Vallabhbhai National Institute of Technology, Surat
Department of Computer Science and Engineering
B Tech II Computer Science and Engineering – Fourth Semester
CS206 – Automata and Formal Languages
End Semester Examination, April 2026

Time: 08:45 AM to 11:45 AM

Date: 28-04-2026

Marks: 100

Instructions:

- Please start the answer to each question on new page ONLY of your answer sheets.
- Please write your correct exam no without fail on the answer sheets as well as the question papers.
- Please be brief and to-the-point in the attempting the answers to the theory questions. Unnecessary and unjustified elaboration will not fetch more marks.
- Bloom's Taxonomy level and Course outcome mapping – Understanding – CO1, Apply – CO2, Analysis – CO3, Evaluate – CO4 and Design – CO5

Q.1	Find a minimum-state Finite Automata recognizing the language corresponding to following regular expressions: (a) $(0^*10 + 1^*0)(01)^*$ (b) $(010)^*1 + (1^*0)^*$	[08] CO1[M] CO2[H] CO5[M]
Q.2	State the pumping lemma theorem for regular language. In each of the following languages defined by regular expressions, prove that L is regular or nonregular using pumping lemma theorem: (a) $L = \{0^n10^{2n} \mid n \geq 0\}$ (b) $L = \{0^i1^j0^k \mid k > i+j\}$ (c) $L = \{0^i1^j \mid j=i \text{ or } j=2i\}$	[09] CO2[M] CO3[H]
Q.3	For each statement below, decide whether it is true or false. If it is true, prove it and if not, give a counter example. All languages are defined over the alphabet $\{0,1\}$. (a) If L is nonregular then L' (complement of L) is nonregular. (b) If L_1 is regular and L_2 is nonregular, then $L_1 \cup L_2$ is nonregular. (c) If L_1 is regular, L_2 is nonregular and $L_1 \cap L_2$ is regular then $L_1 \cup L_2$ is nonregular. (d) if L_1 is regular, L_2 is nonregular and $L_1 \cap L_2$ is nonregular then $L_1 \cup L_2$ is nonregular.	[12] CO2[H] CO4[M]
Q.4 (a)	Defined the document type definition (DTD) for library-book records. Each book-record consists of author-name, title, publisher, year, edition. Author-name may be consists of multiple names. Publisher detail consists of publisher-name and address. The address consists of street-name, town-name, country-name and pin-code. Show the equivalence of DTD with production rule used in CFG.	[05] CO1[M] CO2[M] CO5[H]
Q.4 (b)	Design the following state machines. 1. Moore machine – which outputs residue mod 4 for binary input string. 2. Mealy machine – which increments given input binary number by one. 3. Mealy machine – which recognizes strings accepted by the regular expression $(0+1)^*(00+11)$.	[06] CO2[M] CO5[H]
Q.5	Answer the following:	[12]
(a)	List and define the four types of grammars in the Chomsky hierarchy. Which automaton corresponds to each grammar type in the Chomsky Hierarchy? With suitable example prove that the context free language is <u>not closed under the intersection</u> and complement operation.	CO1[H] CO2[M] CO3[M]
(b)	Design a push down automata which will accepts the language $L = \{a^m b^n c^p d^q \mid m + n = p + q \text{ where } m, n, p, q \geq 1\}$. Design a transition diagram and give seven tuples specification for the PDA. Starting from initial ID (q_0, w, z_0) , show all the reachable ID's when the input w is <u>aabcbdd</u> .	CO1[H] CO2[M] CO4[M] CO5[H]